

NETWORK SCIENCE

Input-driven dynamics for robust memory retrieval in Hopfield networks

Simone Betteti^{1*†}, Giacomo Baggio^{1†}, Francesco Bullo^{2†}, Sandro Zampieri^{1†}

The Hopfield model provides a mathematical framework for understanding the mechanisms of memory storage and retrieval in the human brain. This model has inspired decades of research on learning and retrieval dynamics, capacity estimates, and sequential transitions among memories. Notably, the role of external inputs has been largely underexplored, from their effects on neural dynamics to how they facilitate effective memory retrieval. To bridge this gap, we propose a dynamical system framework in which the external input directly influences the neural synapses and shapes the energy landscape of the Hopfield model. This plasticity-based mechanism provides a clear energetic interpretation of the memory retrieval process and proves effective at correctly classifying mixed inputs. Furthermore, we integrate this model within the framework of modern Hopfield architectures to elucidate how current and past information are combined during the retrieval process. Last, we embed both the classic and the proposed model in an environment disrupted by noise and compare their robustness during memory retrieval.

INTRODUCTION

Since the beginning of the 1980s, the words “associative memory network” have closely echoed with “Hopfield network” (1, 2), and a plethora of subsequent works have endeavored to provide a detailed picture of the properties of such networks (3–5). Drawing from the toolbox of statistical mechanics, Hopfield networks provided a convincing explanation for the multi-stability of memories as function of the neurons couplings and therefore a plausible, dynamic retrieval mechanism over an energy landscape. Recently, in a machine learning–driven Renaissance for associative memory networks, the original framework has been generalized to higher-order interactions (6) and to multilayered architectures (7, 8), thus endowing the model with both a substantially improved capacity (9) and a direct bridge to state-of-the-art transformer models and their attention mechanism (10). Moreover, the framework has paved the way for thought-provoking hypotheses on how neurons and astrocytes could interact (11), at the functional level, to support cognitive processes. The effort to bridge formal approaches and neuroscience is of paramount importance for the advancement of both fields. As proposed in (12), attractor dynamics may be a key component of hippocampal functioning, where the signal relayed by cortical areas is sparsified and orthogonalized in the CA3–CA1 regions (13, 14). In addition, simple attractor models provide a viable tool to study global cortical dynamics in the brain (15, 16), by partitioning the surface in interacting patches of cortex each idealized by Hopfield like networks.

In classic treatments on computational neuroscience (17–19), memory retrieval in the Hopfield model is implicitly described as a two-step process. First, a noisy or incomplete input is presented as a cue and adopted as an initial condition. Then, driven by an energy landscape, the network state flows toward the closest energy minimum representing the prototypical memory.

The literature, however, lacks an explanation for how an external input becomes an initial condition in the neural dynamics; it is worth

noting that external inputs and initial conditions play distinct roles in the behavior of a dynamical system. While this classic two-step process is natural within an algorithmic paradigm, it fails to explain how neuronal circuits continuously react and adapt in real time to external inputs.

In light of these limitations, we advocate for a paradigm shift from a two-step mechanism, akin to a standard algorithmic approach, to an input-driven dynamic mechanism, aligned with the principles of online algorithms and continual learning (20–22). To this extent, we propose a version of the Hopfield model that is driven by external inputs. A key feature of this model is that the input shapes the energy landscape and affects the resulting gradient descent flow (see Fig. 1). Furthermore, our model admits a simple representation as a modern Hopfield network (7, 10); this representation provides a conceptual bridge with the recent literature on transformer models and machine learning. Last, the addition of noise reveals the advantageous integration of past and present information by our model, thereby reducing misclassification errors induced by inconsistent or “glitchy” inputs.

RESULTS

Primer on Hopfield networks

Hopfield networks are a fundamental tool in the study of high level, distributed memories retrieval (1). They simplify neural dynamics into the interplay of two components: a dissipative flow, constantly polarizing the network state toward its resting value, and a synaptic flow, which takes into account the weighted sum of the incoming activity from other neurons in the network. Namely

$$\begin{cases} \dot{\mathbf{x}}(t) = \underbrace{-\mathbf{x}(t)}_{\text{dissipation}} + \underbrace{\mathbf{W}\Psi(\mathbf{x}(t))}_{\text{synaptic interconnections}} \\ \mathbf{x}(0) = \mathbf{x}_0 \in \mathbb{R}^N \end{cases} \quad (1)$$

where the prototypical memories $\{\xi^\mu\}_{\mu=1}^P$, $\xi^\mu \in \{-1, +1\}^N$ are assumed to be orthogonal or with entries that are independent and identically distributed. They are stored in the synaptic matrix $\mathbf{W}$ through one-shot Hebbian learning as

Copyright © 2025 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC).

¹Department of Information Engineering, University of Padua, Padua 35131, Italy.

²Center for Control, Dynamical Systems and Computation, University of California at Santa Barbara, Santa Barbara, CA 93106, USA.

*Corresponding author. Email: bettetisim@dei.unipd.it

†These authors contributed equally to this work.

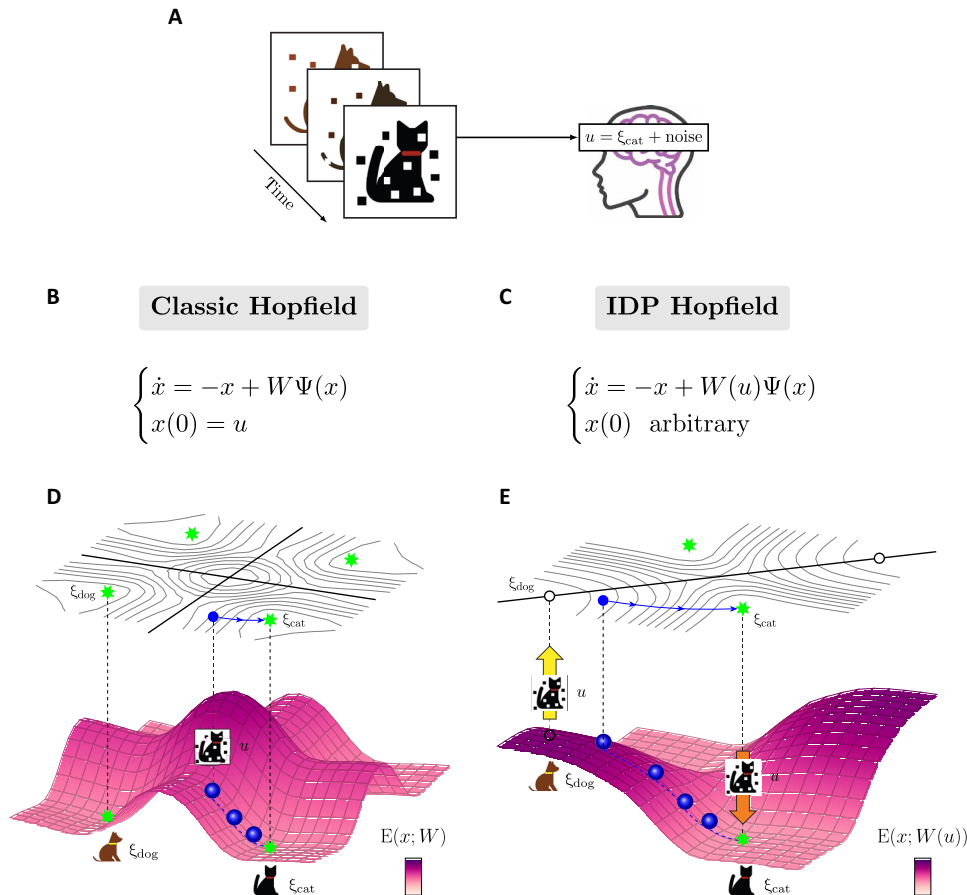


Fig. 1. Comparison between classic Hopfield and IDP Hopfield models. (A) A slowly morphing sequence of noisy images is presented as an input to the observer, who updates its belief state to retrieve the memory closest to the current image $\mathbf{u}$. This adaptation process occurs continuously. (B) In the classic model, the network state is set to an initial condition $\mathbf{x}(0)$ equal to the current image $\mathbf{u}$, and, then, the Hopfield dynamics performs the memory retrieval task. (C) In the proposed input-driven plasticity (IDP) model, the network initial condition is arbitrary, the image $\mathbf{u}$ modifies the synaptic weights $\mathbf{W}(\mathbf{u})$, and the Hopfield dynamics with modified synaptic weights performs the memory retrieval task. This dynamics is well posed and naturally tracks the morphing images also when the image is time-varying $\mathbf{u} = \mathbf{u}(t)$. (D) In the classic model, the Hopfield dynamics is a gradient descent for the energy $E(\mathbf{x}; \mathbf{W})$: The blue ball, representing the neural state, rolls from an initial condition toward a stable minimum point (cat memory). Therefore, the retrieval process is successful when the noisy image $\mathbf{x}(0) = \mathbf{u}$ (dotted cat) lies in the region of attraction of the correct memory (cat memory). (E) In the proposed model, the noisy image $\mathbf{u}$ directly modifies the synaptic matrix $\mathbf{W}(\mathbf{u})$ and, in turn, the energy landscape $E(\mathbf{x}; \mathbf{W}(\mathbf{u}))$, thereby extending the region of attraction of the correct memory. The retrieval process is successful from generic initial conditions when the correct memory is the unique minimum of the landscape. Specifically, in the panel, the noisy image (dotted cat) renders the correct memory a minimum (cat memory) and the incorrect memory (dog memory) no longer an equilibrium of $E(\mathbf{x}; \mathbf{W}(\mathbf{u}))$.

$$\mathbf{W} = \frac{1}{N} \sum_{\mu=1}^P \xi^{\mu} \xi^{\mu\top} \quad (2)$$

Under suitable assumptions on the activation function Ψ (7), convergence to any of the stored memories is guaranteed (2) by the existence of the energy function (see Fig. 1D)

$$E(\mathbf{x}; \mathbf{W}) = -\frac{1}{2} \Psi(\mathbf{x})^{\top} \mathbf{W} \Psi(\mathbf{x}) + \mathbf{x}^{\top} \Psi(\mathbf{x}) - \sum_{i=1}^N \int_0^{x_i} \Psi_i(z) dz \quad (3)$$

given an appropriate initial condition $\mathbf{x}_0$.

The model, first proposed as a discrete time dynamic system (1), has been particularly successful and captivating at explaining pattern reconstruction starting from a partial or corrupted cue (18, 19). In its simplicity, the original theory effectively framed memory retrieval in

the cascade of reactions that lead a network of elementary computational units to fix in a meaningful collective state, i.e., the prototypical memory. Furthermore, the geometric constraints placed on the prototypical memories and on the synaptic matrix allowed for a detailed study of the network capacity. In the original paper and subsequent works (1, 23), it was estimated at around $0.14N$ and, only later, was refined (24) at $\frac{N}{6 \log(N)}$ for exact convergence in probability. These estimates represent the maximal number of memories that can be stored in $\mathbf{W}$ and effectively retrieved as stable equilibria for the Hopfield dynamics.

The original model has then been expanded to the continuous setting (2) with the inclusion of external inputs from the environment, according to the following equation

$$\dot{\mathbf{x}}(t) = -\mathbf{x}(t) + \mathbf{W} \Psi(\mathbf{x}(t)) + \mathbf{u}(t) \quad \mathbf{u}(t) \in \mathbb{R}^N \quad (4)$$

The introduction of an external input conflicts with the previous definition of prototypical memories as stable patterns of activity encoded in $\mathbf{W}$. If the input is time-varying, then the energy function is no longer Lyapunov, namely, it is not decreasing along the system trajectories, and, hence, we may lose convergence to any stable pattern. On the other hand, if the input is constant, then it may distort the energy landscape in such a way that the minima are not related to any of the prototypical memories. When the input is a heterogeneous mixture of memories, the energy landscape is inevitably corrupted and none of the stored memories is retrievable (see Fig. 2A). A possible solution is to clamp the input (25) for a brief time interval δt as an external driver for the dynamics, then un-clamp it, and let the dynamics naturally evolve, namely

$$\dot{\mathbf{x}}(t) = -\mathbf{x}(t) + \mathbf{W}\Psi(\mathbf{x}(t)) + \mathbf{u}(t)\zeta(t) \quad (5)$$

$$\zeta(t) \propto 1_{[0,\delta t]}(t) \quad (6)$$

where $1_S(t)$ is the characteristic function associated to any subset S of the real numbers defined by letting $1_S(t) = 1$ if $t \in S$ and $1_S(t) = 0$

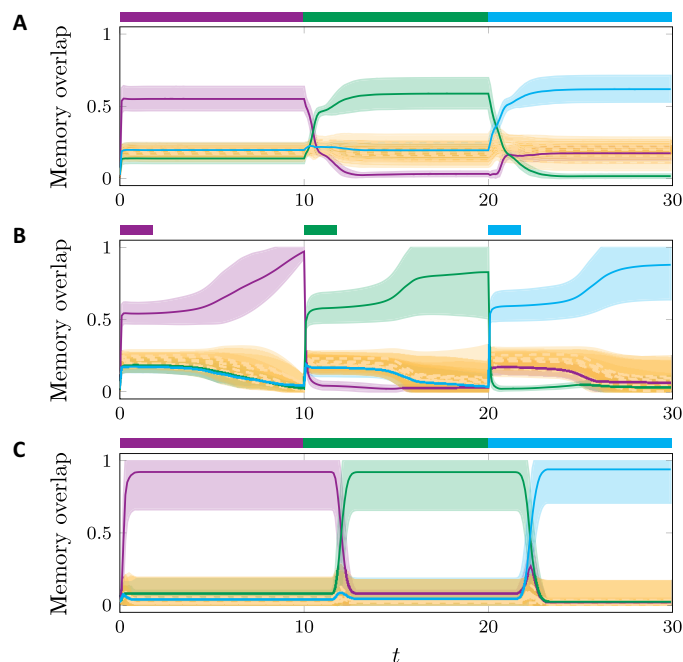


Fig. 2. Exploration of the response of different associative memory models to a time-varying input. Each retrieval plot displays a sample average of the memory overlap parameter over 50 retrieval tasks, with the associated SD shown through the shaded, color-matched surrounding area. (A to C) The response of three different models to a time-varying input. The simulation time is segmented, with each subinterval featuring a constant input. The horizontal bars on top of each retrieval plot display the duration of the external input and the color associated to its dominant saliency weight. At each external input switch, the previous dominant saliency weight shrinks below the stability threshold $\alpha_{\text{stability}}$. (A) Classic Hopfield model (Eq. 4). The network dynamics converge to a mixed state, precluding exact retrieval of any individual memory. (B) Input-modulated Hopfield model (Eq. 5). The modulator (Eq. 6) shuts off the input after intervals of $t = 2$ simulation time. The network dynamics then freely recall the prototypical memory associated with the dominant component in the input. (C) IDP Hopfield model (Eq. 7). The network naturally recalls the prototypical memory associated with the dominant component in the input. The model exhibits memory retention even after input switching, showcasing an intriguing integrative feature for past and present information.

otherwise. The external driver shoots the state trajectory in the direction of the subsequent input, acting as bias for the next retrieval. The model (Eq. 5) is successful at discriminating the dominant component of very mixed inputs (see Fig. 2B), but it does so at the cost of introducing network-wide synchrony. In addition, as observable from Fig. 2 (A and B), each subsequent input instantaneously alters the network activity, canceling the information about any previous activity. Instead, in Fig. 2C, our framework displays a remarkable capability of successfully retrieving the correct memory given the continuous external input, and it will be presented in the next section.

The IDP Hopfield model

The aim of this work is that of studying a plausible mechanism that, given a mixed input, first maintains fixation for short transients on what was previously retrieved, which is past information, and then gradually merges it with the current input, which is present information. This gradual integration should, at a certain point, favor the present information and retrieve the correct memory. We thereby introduce an externally modulated Hopfield model, named input-driven plasticity (IDP) Hopfield model, that tries to capture this exact phenomenology. The IDP Hopfield model is defined as

$$\dot{\mathbf{x}}(t) = -\mathbf{x}(t) + \mathbf{W}(\mathbf{u}(t))\Psi(\mathbf{x}(t)) \quad (7)$$

where for a piecewise continuous, bounded input $\mathbf{u}(t) \in \mathbb{R}^N$, the input modulated synaptic matrix is

$$\mathbf{W}(\mathbf{u}(t)) = \frac{1}{N} \sum_{\mu=1}^P \alpha_{\mu}(t) \xi^{\mu} \xi^{\mu T} \quad (8)$$

where $\alpha_{\mu}(t) := [\xi^{\mu T} \mathbf{u}(t)]^2$ are called the saliency weights, in accordance with previous literature (26, 27). These previous studies have explored the role of saliency weights and their effect on the dynamics (Eq. 7) within the context of the morphing problem, focusing on how these weights drive transitions between memories as they evolve. However, a comprehensive analysis of how the magnitude of saliency weights affects the existence and stability of memories as equilibrium points remains missing. To address this gap, we begin by characterizing the input, followed by a detailed examination of the specific conditions on saliency weights necessary to ensure the existence and stability of memories as equilibria for Eq. 7. We say that an input is homogeneous when it is substantially aligned with one of the prototypical memories and almost orthogonal to the others, i.e., when there exists a $\nu = 1, \dots, P$ such that $\alpha_{\nu} \gg 0$ and $\alpha_{\mu} \approx 0$ for all $\mu \neq \nu$. Conversely, we say that the input is well mixed when it has similar saliency weights associated to all the prototypical memories, i.e., $\alpha_{\mu} \approx \alpha_{\nu}$ for all μ and ν . Furthermore, under a timescale separation hypothesis for neural dynamics and biologically trackable inputs, i.e., under the hypothesis that neural dynamics are much faster than input dynamics, the latter can be considered constant throughout each retrieval time interval. Unlike Hopfield's foundational works based upon a static synaptic matrix, we build on insights from (26, 28, 29) to introduce elementary dynamics into the synaptic structure. Differently from these treatments, we assume the synaptic changes to depend on an externally generated input rather than on a feedback activity of the associative memory system, such as that coming from intra-network astrocytes. This dynamic approach yields advantageous properties for system behavior, particularly by demonstrating how externally modulated synaptic changes can promote the retrieval of certain patterns while inhibiting others. To rigorously quantify

how the input determines specific retrieval properties at the network level, we analytically study the existence and stability of memories, specifically for the case of monotonically increasing, odd activation functions $\Psi(\mathbf{x}) = (\psi(\mathbf{x}_1), \dots, \psi(\mathbf{x}_N))$. A full characterization of the assumptions and the proofs for the presented results are presented in the Supplementary Materials.

1) Given a prototypical memory ξ^μ , the associated retrievable memory is a vector of the form $\mathbf{x}_\mu = \gamma_\mu \xi^\mu$ for $\gamma_\mu \neq 0$ and is an equilibrium of Eq. 7.

2) A retrievable memory exists under the current input if and only if the associated saliency weight is larger than an existence threshold

$$\alpha_\mu > \alpha_{\text{existence}} = 1 \quad (9)$$

and in this case γ_μ satisfies

$$\frac{\gamma_\mu}{\alpha_\mu} = \psi(\gamma_\mu) \quad (10)$$

3) If multiple retrievable memories exist under the current input, then there exists a stability threshold $\alpha_{\text{stability}} > 1$ such that each memory $\mathbf{x}_\mu$ is stable if and only if $\alpha_\mu > \alpha_{\text{stability}}$.

Notice that the classic Hopfield model is a specific sub-case of the IDP model, where the input impinging on the synaptic matrix is characterized by $\alpha_\mu \equiv k > 1$, for all μ . As clearly observable from Fig. 3, the external input induces a smooth reshaping of the energy landscape. In particular, the modified energy function is

$$E(\mathbf{x}; \mathbf{W}(\mathbf{u})) = -\frac{1}{2} \Psi(\mathbf{x})^\top \mathbf{W}(\mathbf{u}) \Psi(\mathbf{x}) + \mathbf{x}^\top \Psi(\mathbf{x}) - \sum_{i=1}^N \int_0^{x_i} \psi(z) dz \quad (11)$$

and, for every $\nu \in \{1, \dots, P\}$ such that $\alpha_\nu < \alpha_\mu$, we have that

$$E(\mathbf{x}_\mu, \mathbf{W}(\mathbf{u})) < E(\mathbf{x}_\nu, \mathbf{W}(\mathbf{u})) \quad (12)$$

Hence, the greater are the saliency weights, the deeper the associated energy minima. Exploiting the timescale separation hypothesis for neural dynamics and input dynamics, we also have that

$$\begin{aligned} \frac{d}{dt} E(\mathbf{x}(t); \mathbf{W}(\mathbf{u}(t))) &= \frac{\partial E}{\partial \mathbf{x}}(\mathbf{x}(t); \mathbf{W}(\mathbf{u}(t))) \dot{\mathbf{x}}(t) \\ &+ \underbrace{\frac{\partial E}{\partial \mathbf{u}}(\mathbf{x}(t); \mathbf{W}(\mathbf{u}(t))) \dot{\mathbf{u}}(t)}_{\approx 0} \\ &\approx \frac{\partial E}{\partial \mathbf{x}}(\mathbf{x}(t); \mathbf{W}(\mathbf{u})) \dot{\mathbf{x}}(t) \leq 0 \end{aligned} \quad (13)$$

The IDP Hopfield model naturally endows us with a direct interpretation of when the input is clearly understandable and when instead is too vague to evoke any kind of retrieval. As previously presented, memories exist (are retrievable) only if their respective saliency weights in the input are at least unitary. Thus, for any input that is well mixed and such that $\alpha_\mu < 1$ for all μ (Fig. 3A), the dynamics will converge to the origin, which we call confusion state. This state exists only in the IDP model, as the classic model always has all the retrievable memories irrespective of the input. The possibility of falling into a confusion state on the basis of what is experienced seems quite plausible, as the clarity of what we sense in our everyday experience has a clear bearing on what we recall. Last, we want to address the following question: Does the initial condition still matter?

The energy function $E(\mathbf{x}; \mathbf{W})$ ensures convergence to one of the stable memories from any initial conditions. On the other hand, the stability threshold $\alpha_{\text{stability}}$ determines how many stable memories we have for a given input $\mathbf{u}$. Thus, the fewer the stable memories for the IDP Hopfield model, the wider their basin of attraction. In the extremal case of a single stable memory, the whole $\mathbb{R}^N$ is its basin of attraction, and any initial condition leads to the correct retrieval. Thus, appropriately initializing the system becomes irrelevant, especially in the presence of perturbations that push the dynamics out of shallow energy minima (see Fig. 3D).

A modern interpretation

As we mentioned in the introduction, the IDP model naturally lends itself to a description through the modern Hopfield formalism outlined in (7). By means of this formalism, a recurrent neural network such as the Hopfield model can be effectively deconstructed into two interacting layers void of intralayer connections.

The following tripartite architecture characterizes a more general model that captures the interaction of the activity in the memory layer $\mathbf{y} \in \mathbb{R}^P$ and in the saliency layer $\mathbf{z} \in \mathbb{R}^P$. The combined information is then exploited to drive the retrieval in the feature layer $\mathbf{x} \in \mathbb{R}^N$. In summary, the modern Hopfield reformulation of the IDP model is

$$\tau_x \dot{\mathbf{x}}(t) = -\mathbf{x}(t) + \mathbf{M}_x [\mathbf{z}(t) \odot \Psi_x(\mathbf{z}(t) \odot \mathbf{y}(t))] \quad (14)$$

$$\tau_y \dot{\mathbf{y}}(t) = -\mathbf{y}(t) + \mathbf{M}_y \Psi_y(\mathbf{x}(t)) \quad (15)$$

$$\tau_z \dot{\mathbf{z}}(t) = -\mathbf{z}(t) + \mathbf{M}_z \Psi_z(\mathbf{u}(t)) \quad (16)$$

where the symbol $\odot$ is the Hadamard entrywise product, namely, $(\mathbf{z}(t) \odot \mathbf{y}(t))_i = y_i(t) z_i(t)$, and Ψ_z is an activation function implementing either a linear or nonlinear processing of the input. Notice that, when $\mathbf{M}_x = \mathbf{M}$ and $\mathbf{M}_y = \mathbf{M}_z = \mathbf{M}^\top$ with $\mathbf{M} = N^{-\frac{1}{2}} [\xi^1 \dots \xi^P] \in N^{-1/2} \{-1, 1\}^{N \times P}$, and Ψ_x, Ψ_z are identity functions, Eqs. 14 to 16 reduce to the IDP Hopfield model (Eq. 7) in the limits $\tau_y \rightarrow 0$ and $\tau_z \rightarrow 0$, with $\alpha(t) = \mathbf{z}(t) \odot \mathbf{z}(t)$. Unraveling the activity of the IDP Hopfield network into the distinct components allows for a better qualitative understanding of the layers' contribution (see Fig. 4 for a block representation). The memory layer serves the function of pooling layer, projecting the activity of the feature layer into a similarity space. The pooled activity is then modulated by the input decomposition. It is clear then that the combination of pooling and modulation implements a natural trade-off between past internal activity and externally incoming information.

In the model discussed, so far, we have the input processing $\mathbf{M}_z \Psi_z(\mathbf{u}(t)) = \mathbf{M}^\top \mathbf{u}(t)$, because it allows a direct logical and visual interpretation of the relationship between the input decomposition and the network response. However, in a realistic neuronal system, the relationship may not be so simple and interpretable. The most general case would be that where $\mathbf{M}_z \in \mathbb{R}^{P \times K}$ and $\mathbf{u}(t) \in \mathbb{R}^K$ is a generic input. This is the case when the input originates from a different cortical region possibly having a different output dimensionality, and, therefore, the relation with the prototypical memories is more subtle. A discussion of how the Ψ_z output channels are related to the memories stored in the recurrent network is beyond the scope of this article but remains an important problem open to original solutions.

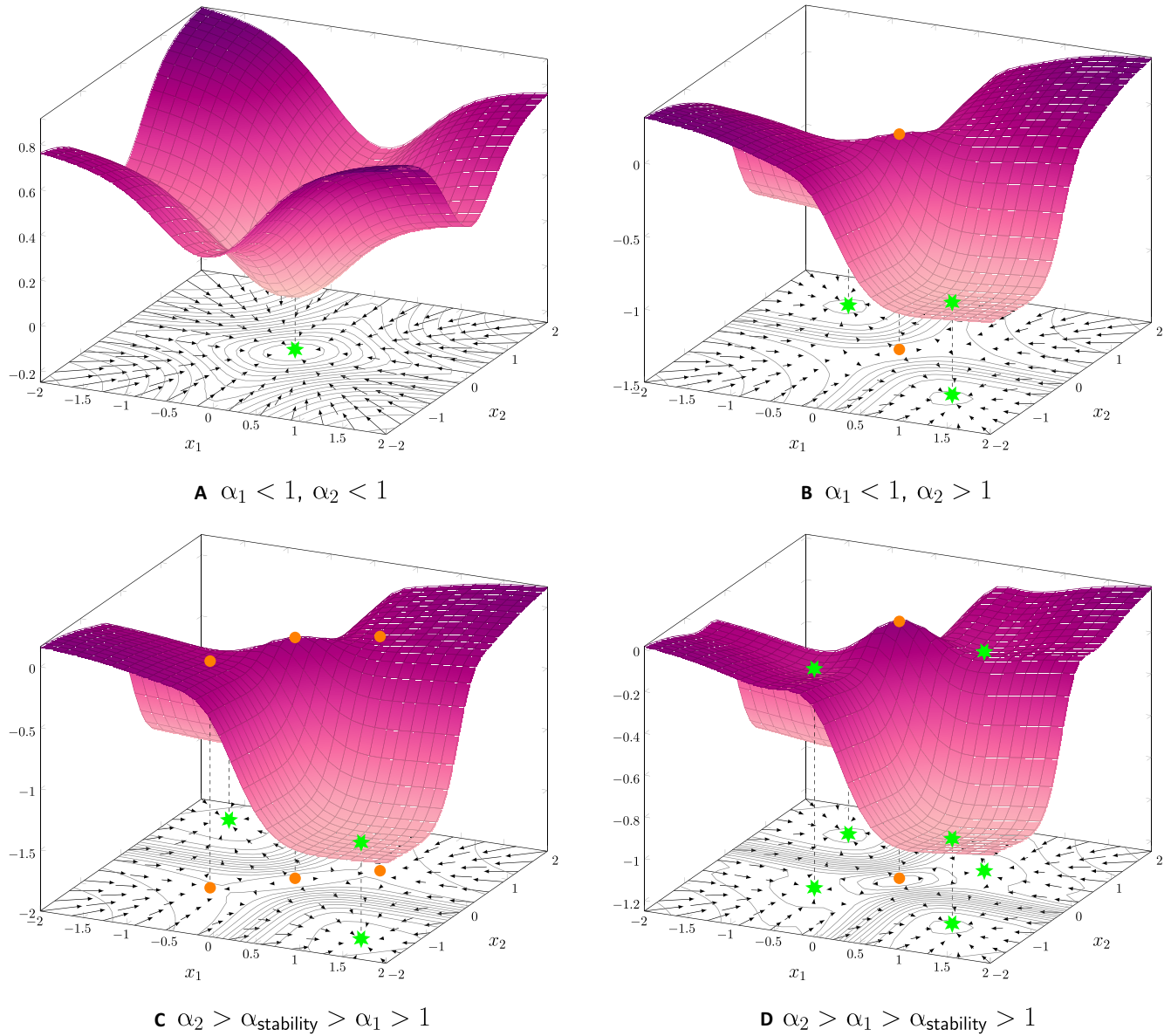


Fig. 3. Energy landscapes for IDP Hopfield model for varying saliency weights. Stable and unstable equilibria are depicted as green stars and orange dots, respectively. Recall the existence threshold is $\alpha_{\text{existence}} = 1$ and, when multiple memories exist in the input, the stability threshold satisfies $\alpha_{\text{stability}} > 1$. **(A)** “No memories” $\alpha_1 < 1, \alpha_2 < 1$: When no memory is sufficiently strong in the input, the only global minimum is at the origin, and it is globally attractive for the dynamics. This situation corresponds to a confusion state for the network, in which the input is not strong enough to evoke any retrieval. **(B)** “One memory” $\alpha_1 < 1, \alpha_2 > 1$: When the saliency weight for precisely one memory is above $\alpha_{\text{existence}}$, two symmetric equilibria appear corresponding to the memory, and they are attractive from almost all initial conditions. In this case, the origin is a saddle point. **(C)** “one stable and one unstable memory” $\alpha_2 > \alpha_{\text{stability}} > \alpha_1 > 1$: Two symmetric equilibria appear corresponding to the stable memory, and they are attractive from almost all initial conditions. The other two symmetric equilibria are saddle points and the origin becomes an unstable maximum. **(D)** “Two stable memories” $\alpha_2 > \alpha_1 > \alpha_{\text{stability}}$: Four symmetric stable equilibria appear, corresponding to the two memories of the model. The memory associated to the dominant saliency weight carves deeper valley in the energy landscape than the other memory. When $\alpha_1 \approx \alpha_{\text{stability}}$, the shallowness of the valley associated to the first memory facilitates outward jumps due to stochastic fluctuations.

Last, the framework of Eqs. 14 to 16 provides a dynamic description of a biased self-attention mechanism in a transformer, generalizing the results of (10). Choosing $\Psi_y(\cdot) = \text{Id}(\cdot)$, $\Psi_x(\cdot) = \text{softmax}(\cdot)$, and $\mathbf{M}_x = \mathbf{M}_y^\top = \mathbf{M}$ and taking $\tau_x = 1$, $\tau_y \rightarrow 0$, and $\tau_z \rightarrow 0$ yield

$$\dot{\mathbf{x}} = -\mathbf{x} + \mathbf{M}[\mathbf{z} \odot \text{softmax}(\mathbf{z} \odot \mathbf{M}^\top \mathbf{x})] = -\nabla_{\mathbf{x}} E_{tr}(\mathbf{x}; \mathbf{z}) \quad (17)$$

$$E_{tr}(\mathbf{x}; \mathbf{z}) = -\frac{1}{\beta} \log \left(\sum_{i=1}^N e^{\beta \mathbf{z}_i (\mathbf{M}^\top \mathbf{x})_i} \right) + \frac{\mathbf{x}^\top \mathbf{x}}{2} \beta > 0 \quad (18)$$

Constraining $\mathbf{z}(t) \equiv \mathbf{1}_N$, we recover exactly the dynamic self-attention mechanisms studied in the provided references. Differently

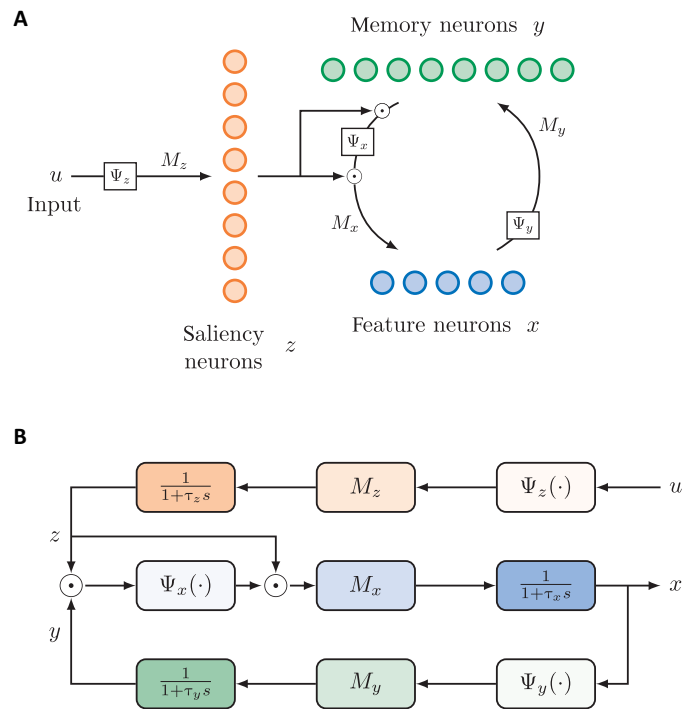


Fig. 4. Illustration of the modern Hopfield reformulation of the dynamics for the IDP Hopfield model with input filter. The symbol $\odot$ denotes a Hadamard entry-wise product. (A) Neural network representation with interconnected layers and synaptic weights M_z , M_x , and M_y . (B) Block diagram representation, where each block $\frac{1}{1+\tau_x s}$, with $\tau = \tau_z, \tau_x, \tau_y$, denotes a low-pass filter with cutoff frequency $1/\tau$, each block M_z , M_x , and M_y denotes a matrix-vector multiplication, and each block $\Psi_z(\cdot)$, $\Psi_x(\cdot)$, and $\Psi_y(\cdot)$ denotes an activation function.

from the standard implementation of the attention mechanism, augmenting the dynamics with Eq. 16 may help in controlling the retrieval of keys as a function of continuously streamed external biases pre-multiplying the queries, hence in a more online fashion.

IDP induces resiliency to noise

Both the classic (Eq. 5) and the IDP Hopfield (Eq. 7) models are fully capable of retrieving the correct memory when subjected to the same well-mixed input (see Fig. 2, B and C). This may mislead the reader to conclude that the two models are equivalent. However, both are still ideal models, where neural units are perfectly insulated. Thus, they fail to capture the essence of real neural phenomena in settings with ubiquitous background noise. Thus, we consider the associated stochastic dynamics

$$d\mathbf{x}(t) = \mathcal{H}(\mathbf{x}(t)) dt + \sigma d\mathbf{B}(t) \quad (19)$$

where the drift term $\mathcal{H}$ is in place of a generic field chosen between the classic and IDP models, $\sigma \geq 0$ is a diffusion term that will henceforth be called noise amplitude, and $\mathbf{B}(t)$ is the standard n -dimensional Brownian motion.

From Fig. 5A, observe that, when the system is insulated ($\sigma = 0$) and for highly mixed inputs, the retrieval process remains stuck on a previous memory as long as the associated saliency weight is above the stability threshold $\alpha_{\text{stability}}$. Equivalently, the dynamics are entrapped in a shallow minimum of the energy $E(\mathbf{x}; \mathbf{W}(\mathbf{u}))$. This behavior is undesirable, as it impairs the system's ability to correctly

regulate its response under conditions of high input ambiguity and, thereby, undermine its reliability. We here argue that the IDP model is not only robust to noise but actually benefits from and exploits it as a mean to escape the shallow minima and to achieve efficient retrieval in vague contexts. As observable from Fig. 5B, relatively high amplitude noise ($\sigma = 4$) favors the consecutive retrieval of different memories even when the inputs are highly mixed. Consistent with our intuition in Fig. 3D, the noise drives the retrieval process out of shallow energy valleys and into the deepest energy minima. Thus, the stochastic IDP model embodies a mechanism that actively prioritizes the most relevant features in its environment while suppressing the processing of background elements, mirroring the psychological phenomenon of selective attention (30). To promote a fair comparison, we are interested in exploring the same response to noise in the classic Hopfield model. Sampling $\sigma \in [0, 2]$, we observe from Fig. 5C that even moderate noise amplitudes ($\sigma = 1$) disrupt retrieval, causing overlaps to remain at chance level. Further increasing σ leads to total retrieval failure, indicating that the classical Hopfield model exhibits limited robustness to stochastic perturbations.

The observed robustness to noise for the IDP Hopfield model entails the existence of a range of noise amplitudes σ , leading to correct sequential retrieval. It is then natural to wonder whether there exists optimal value of σ that minimize the time elapsing between input onset and the next retrieval. As observable from Fig. 5F, for an insulated system ($\sigma = 0$), the time elapsing between input onset and the decay of the past memory retrieval curve is quite long, but, once initiated, the transient is very fast. Despite the successful switching, the prolonged dwelling on the previous memory seems somewhat undesirable for systems that have evolved to efficiently and readily adapt to their environment. When the system has relatively high noise amplitude ($\sigma = 4$), the decay of the past memory retrieval curve is very swift, but the transient toward the next retrieval is fairly slow. Slow tracking of a change in the input seems equally undesirable from an adaptive perspective. Instead, for intermediate values of the amplitude parameter ($\sigma = 2$), the model displays both a relatively fast decay of the past memory retrieval curve and a swift transient toward the next retrieval. In particular, it can be observed that intermediate noise amplitudes result in shorter times between input onset and retrieval of the correct memory with respect to both insulated and highly noisy systems.

In addition, brief prioritization of past information following a change in the input allows the system to correct transient errors or glitches. For desirable values of the noise amplitude σ , the IDP model adequately weights the importance of past information, so that glitches are neglected. To test this hypothesis, we conduct a numerical experiment where the dominant saliency weight undergoes a brief perturbation (see the green glitch at $t = 8$ in Fig. 5D). The IDP Hopfield model effectively suppresses these transient fluctuations, maintaining retrieval stability. This robustness arises from the multiplicative mechanism in Eq. 14, which prevents abrupt disruptions in retrieval dynamics by integrating present external information against past feedback activity. Instead, the classical Hopfield model (see Fig. 5E) fails to correct the glitch, with instantaneous transients aligned with the provided input. Furthermore, the memory overlap trajectories for the classical Hopfield model quickly decay to zero when the input is switched off.

The presented results may additionally benefit from a more rigorous study of the properties of the stochastic Hopfield model. A

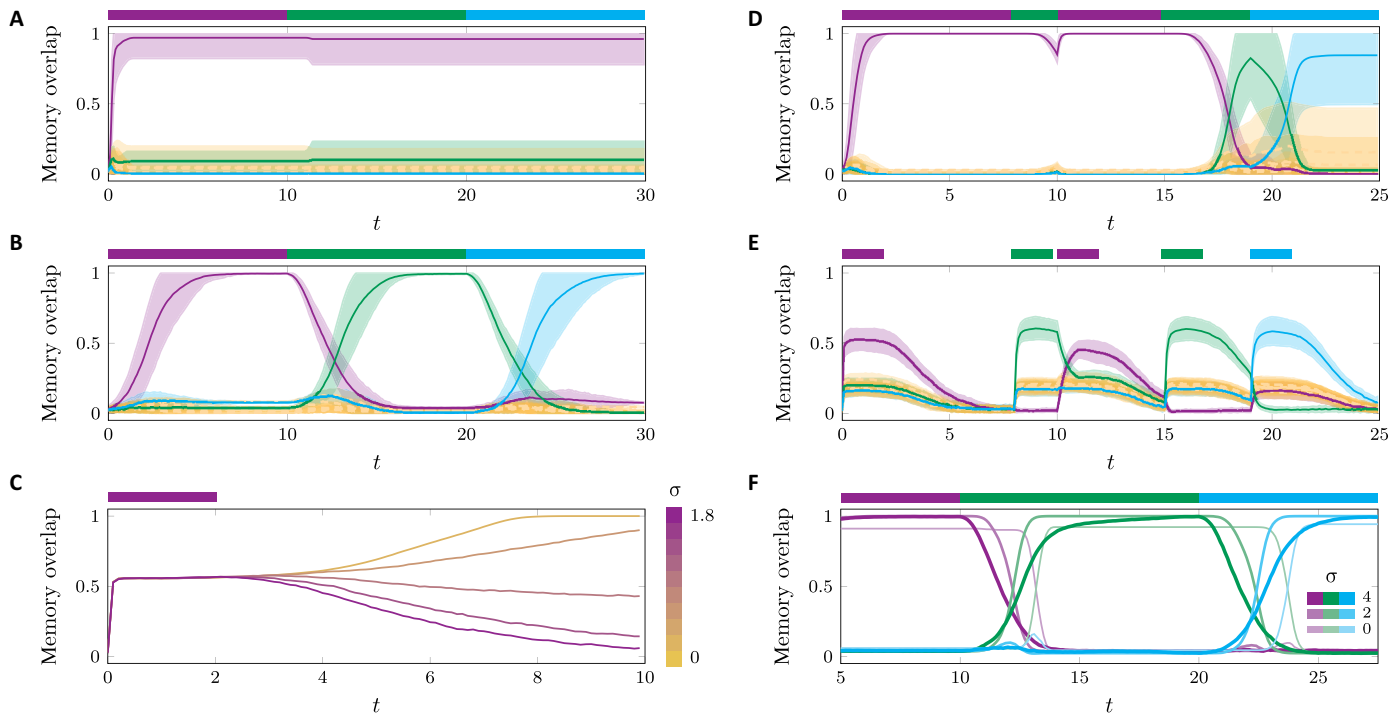


Fig. 5. Robustness to noise in the IDP and classic Hopfield model. (A and B) Differently from Fig. 2C, the prototypical memory associated to the red weight remains stable after each input switching. (A) IDP model insulated from noise. The model correctly retrieves the first prototypical memory but is unable to track the dominant component of next inputs due to persistent stability of the red component. (B) IDP model with high amplitude noise. The model correctly retrieves the first prototypical memory. At input switch, the memory remains stable, but the associated energy well considerably flattens (see Fig. 3D as reference case). The noise pushes the activity out of the flat wells and directs it toward the deepest one, which is associated to the dominant saliency weight. The process is then repeated at each input switch. (C) The classic Hopfield model preserves good memory retrieval performance only for small noise amplitudes. As the noise amplitude increases, the memory retrieval performance quickly deteriorates until total failure. (D and E) Robustness to brief input perturbations (glitches). (D) The IDP Hopfield model successfully corrects the glitch integrating past internal and present external information. (E) The classic Hopfield model fails at correcting the glitch and fails in the retrieval task once the input is switched off. (F) Noise amplitude and speed of the transient toward the next memory. Intermediate values of noise amplitude result in optimal speed for the transition between two memories. Insulated and high amplitude noise models take considerably longer to complete the switch.

valuable tool toward this aim is the Fokker-Planck equation (31–33), which describes the time evolution of the probability measure associated to Eq. 19. Wong (34) studies a particular case of the stochastic Hopfield model, where the associated probability measure converges to a stationary distribution that explicitly depends on the energy $E(\mathbf{x}; \mathbf{W})$, revealing that probability mass concentrates around its minima. In the supplementary information, we discuss a more general case and point out existing limitations and challenges in the analysis of the Fokker-Planck equation associated to the Hopfield model.

DISCUSSION

Achieving robust behavior in multistable systems

Multistability is a distinctive feature of the classical Hopfield model, required for its associative memory functionality. Yet, multistability is typically avoided in controlled engineering systems because it potentially leads to fragile and unpredictable behavior. While complex systems, such as the electric power grid, may exhibit multiple stability regions, only one of them corresponds to proper grid functioning. It is the responsibility of the grid control infrastructure to steer the system away from undesired stability regions. The proposed IDP Hopfield model fulfills this same responsibility in the context of

memory retrieval: When the input is not ambiguous, the neural state is reliably driven to the correct stability region, achieving a robust and predictable memory system.

Energy shaping in the IDP Hopfield model

The IDP Hopfield model presents a simple yet effective explanation of how a direct input-driven modulation of the synapses can enrich the dynamic range of recurrent neural networks. The input-driven adjustments of the synaptic couplings between neurons enforce a clear memory hierarchy, with single memories existing only if sufficiently stimulated. Furthermore, the input decomposition changes the stability properties of single memory patterns. Through the existence of the threshold $\alpha_{\text{stability}}$, a memory that was stable at a certain time instant can suddenly become a saddle point and thus allow the network dynamics to roll toward another memory, as proposed by Karuvally *et al.* (35). During the retrieval process, the saliency attributed to individual memories via input decomposition reshapes the energy landscape of the model. This process deepens the wells associated with the most dominant input components while flattening the others. Consequently, the presence of exogenous noise is enough to drive the dynamics toward the deepest basin of attraction and confine the network activity within it. It is worth mentioning

that a model similar to the IDP Hopfield has been recently numerically studied in (36) with the aim of implementing sequential memory retrieval. In this context, the dynamics and distribution of the saliency weights reflect some previous association among prototypical memories and confine the network activity to limit cycles.

Future work and implications

The present study lays the foundation for future research aimed at fully analyzing the biologically plausible firing rate version of the IDP model, a preliminary version of which is outlined in the Supplementary Materials. This study also opens pathways for the empirical validation of input-dependent associative memory models and a comprehensive characterization of learning dynamics, encompassing both long-term memory formation and short-term modulation of saliency weights. Specifically, future works will need to consider a complete theoretical investigation of the IDP firing rate model in a low neural activity regime compatible with experimental evidence (37). A thorough understanding of the dynamics underlying memory retrieval in the firing rate model may guide experimental researchers in their inquiries, and successful validation of the model may provide a solid ground for the understanding of high level cognitive processes. Our formalization of the IDP Hopfield model paves the way for a thorough investigation into its underlying learning dynamics. Developing an effective, biologically plausible learning mechanism, such as one grounded in the Hebbian paradigm, could position the IDP Hopfield model, rooted in the modern Hopfield architecture, as an efficient biologically inspired machine learning model.

In addition, jointly solving the problem of long-term learning and short-term modulations may provide valuable answers on the multiplicity of timescales observed throughout biological learning processes. Notably, there is already ample evidence (38) that heterogeneity in functional features and timescales may be the biological solution to the problems of catastrophic forgetting and adaptive learning. The advancement of such comprehension holds relevant implications, extending its impact beyond neuroscience and onto the domain of machine learning. Within the machine learning field, the mentioned problems prevent virtual and embedded agents (21) from deployment in contexts that require both cognitive flexibility and preservation of past knowledge. The successful combination of these two elements epitomizes the ultimate aspiration of continual learning (22, 39), an emergent machine learning paradigm that seeks to meaningfully integrate present and past data and yields the potential for great technological advancement.

MATERIALS AND METHODS

The simulations detailed in Figs. 2 and 5 use networks of $N = 1024$ neurons and the standard Euler-Maruyama scheme for the integration of the dynamics (Eq. 19) (for $\sigma = 0$, this become the standard forward Euler scheme). The synaptic input to each neuron is then processed by the activation function $\psi(\mathbf{x}_i) = \tanh(5\mathbf{x}_i)$. The external inputs $\mathbf{u}(t)$ used in the simulation are piecewise constant, and they are ad hoc constructed to achieve experimental control over the memory that the Hopfield network should retrieve within a specific time interval. In particular, within each interval of length $t = 10$ of simulation time, we sample the dominant saliency weight $\alpha_\nu \sim \mathcal{U}[7, 10]$ for $\nu = 1, \dots, P$ and all the others as $\alpha_\mu \sim \mathcal{U}[2, 5]$, with $\mu \neq \nu$. The input within each of the defined time interval is of

the form $\mathbf{u} = \sum_{\mu=1}^P \omega_\mu \xi^\mu$, where $\omega_\mu = \sqrt{\alpha_\mu}$. The interested reader is referred to the “Materials and methods - More details” section in the Supplementary Materials for finer-tuned simulation choices.

Supplementary Materials

This PDF file includes:

Supplementary Materials and Methods
Supplementary Text
Figs. S1 to S6
References

REFERENCES AND NOTES

1. J. J. Hopfield, Neural networks and physical systems with emergent collective computational abilities. *Proc. Natl. Acad. Sci. U.S.A.* **79**, 2554–2558 (1982).
2. J. J. Hopfield, Neurons with graded response have collective computational properties like those of two-state neurons. *Proc. Natl. Acad. Sci. U.S.A.* **81**, 3088–3092 (1984).
3. D. J. Amit, H. Gutfreund, H. Sompolinsky, Statistical mechanics of neural networks near saturation. *Ann. Phys. Rehabil. Med.* **173**, 30–67 (1987).
4. A. Crisanti, D. J. Amit, H. Gutfreund, Saturation level of the Hopfield model for neural network. *EPL* **2**, 337–341 (1986).
5. A. Treves, D. Amit, Metastable states in asymmetrically diluted Hopfield networks. *J. Phys. A Math. Gen.* **21**, 3155–3169 (1988).
6. D. Krotov, J. Hopfield, Dense associative memory for pattern recognition. *arXiv.1606.01164*. (2016).
7. D. Krotov, J. Hopfield, Large associative memory problem in neurobiology and machine learning. *arXiv:2008.06996* (2020).
8. H. T. Chaudhry, J. A. Zavattone-Veth, D. Krotov, C. Pehlevan, “Long sequence Hopfield memory” in *Proceedings of the 37th International Conference on Neural Information Processing Systems (NeurIPS, 2023)*, pp. 54300–54340.
9. M. Demircigil, J. Heusel, M. Löwe, S. Uppgang, F. Vermet, On a model of associative memory with huge storage capacity. *J. Stat. Phys.* **168**, 288–299 (2017).
10. H. Ramsauer, B. Schäßl, J. Lehner, P. Seidl, M. Widrich, T. Adler, L. Gruber, M. Holzleitner, M. Pavlović, G. K. Sandve, V. Greiff, D. Kreil, M. Kopp, G. Klambauer, J. Brandstetter, S. Hochreiter, “Hopfield networks is all you need” in *International Conference on Learning Representations (ICLR, 2021)*, pp. 1–95.
11. L. Kozachkov, K. V. Kastanenko, D. Krotov, Building transformers from neurons and astrocytes. *Proc. Natl. Acad. Sci. U.S.A.* **120**, e2219150120 (2023).
12. A. Treves, E. T. Rolls, Computational constraints suggest the need for two distinct input systems to the hippocampal CA3 network. *Hippocampus* **2**, 189–199 (1992).
13. M. Yassa, C. E. Stark, Pattern separation in the hippocampus. *Trends Neurosci.* **34**, 515–525 (2011).
14. E. Rolls, The mechanisms for pattern completion and pattern separation in the hippocampus. *Front. Syst. Neurosci.* **7**, 74 (2013).
15. E. Russo, A. Treves, Cortical free-association dynamics: Distinct phases of a latching network. *Phys. Rev. E Stat. Nonlin. Soft Matter. Phys.* **85**, 051920 (2012).
16. M. Naim, V. Boboeva, C. Kang, A. Treves, Reducing a cortical network to a Potts model yields storage capacity estimates. *J. Stat. Mech.* **2018**, 043304 (2018).
17. D. J. Amit, *Modeling Brain Function: The World of Attractor Neural Networks* (Cambridge Univ. Press, 2012).
18. P. Dayan, L. F. Abbott, *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems* (MIT Press, 2005).
19. W. Gerstner, W. M. Kistler, R. Naud, L. Paninski, *Neuronal Dynamics: From Single Neurons to Networks and Models of Cognition* (Cambridge Univ. Press, 2014).
20. F. Zenke, B. Poole, S. Ganguli, “Continual learning through synaptic intelligence” in *International Conference on Machine Learning (PMLR, 2017)*, pp. 3987–3995.
21. R. Hadsell, D. Rao, A. A. Rusu, R. Pascanu, Embracing change: Continual learning in deep neural networks. *Trends Cogn. Sci.* **24**, 1028–1040 (2020).
22. T. Lesort, V. Lomonaco, A. Stoian, D. Maltoni, D. Filliat, N. Díaz-Rodríguez, Continual learning for robotics: Definition, framework, learning strategies, opportunities and challenges. *Inf. Fusion* **58**, 52–68 (2020).
23. D. J. Amit, H. Gutfreund, H. Sompolinsky, Information storage in neural networks with low levels of activity. *Phys. Rev. A* **35**, 2293–2303 (1987).
24. D. Petritis, *Thermodynamic Formalism of Neural Computing* (Springer, 1996), pp. 81–146.
25. F. P. Battaglia, A. Treves, Stable and rapid recurrent processing in realistic autoassociative memories. *Neural Comput.* **10**, 431–450 (1998).
26. B. Blumenfeld, S. Preminger, D. Sagi, M. Tsodyks, Dynamics of memory representations in networks with novelty-facilitated synaptic plasticity. *Neuron* **52**, 383–394 (2006).

27. H. Tang, H. Li, R. Yan, Memory dynamics in attractor networks with saliency weights. *Neural Comput.* **22**, 1899–1926 (2010).
28. G. Mongillo, O. Barak, M. Tsodyks, Synaptic theory of working memory. *Science* **319**, 1543–1546 (2008).
29. L. Kozachkov, J.-J. Slotine, D. Krotov, Neuron-astrocyte associative memory. arXiv:2311.08135v2 (2023).
30. J. Driver, A selective review of selective attention research from the past century. *Br. J. Psychol.* **92**, 53–78 (2001).
31. H. Yan, L. Zhao, L. Hu, X. Wang, E. Wang, J. Wang, Nonequilibrium landscape theory of neural networks. *Proc. Natl. Acad. Sci. U.S.A.* **110**, E4185–E4194 (2013).
32. B. A. W. Brinkman, H. Yan, A. Maffei, I. M. Park, A. Fontanini, J. Wang, G. la Camera, Metastable dynamics of neural circuits and networks. *Appl. Phys. Rev.* **9**, 011313 (2022).
33. F. Moss, P. McClintock, *Noise in Nonlinear Dynamical Systems* (Cambridge Univ. Press, 2012).
34. E. Wong, Stochastic neural networks. *Algorithmica* **6**, 466–478 (1991).
35. A. Karuvally, T. Sejnowski, H. T. Siegelmann, “General sequential episodic memory model” in *International Conference on Machine Learning* (PMLR, 2023), pp. 15900–15910.
36. L. Herron, P. Sartori, B. Xue, Robust retrieval of dynamic sequences through interaction modulation. *PRX Life* **1**, 023012 (2023).
37. C. Gastaldi, T. Schwalger, E. De Falco, R. Q. Quiroga, W. Gerstner, When shared concept cells support associations: Theory of overlapping memory engrams. *PLOS Comput. Biol.* **17**, e1009691 (2021).
38. M. K. Benna, S. Fusi, Computational principles of synaptic memory consolidation. *Nat. Neurosci.* **19**, 1697–1706 (2016).
39. D. Kudithipudi, M. Aguilar-Simon, J. Babb, M. Bazhenov, D. Blackiston, J. Bongard, A. P. Brna, S. Chakravarthy Raja, N. Cheney, J. Clune, A. Daram, S. Fusi, P. Helfer, L. Kay, N. Ketz, Z. Kira, S. Kolouri, J. L. Krichmar, S. Kriegman, M. Levin, S. Madireddy, S. Manicka, A. Marjaninejad, B. McNaughton, R. Miikkulainen, Z. Navratilova, T. Pandit, A. Parker, P. K. Pilly, S. Risi, T. J. Sejnowski, A. Soltoggio, N. Soares, A. S. Tolia, D. Urbina-Meléndez, F. J. Valero-Cuevas, G. M. van de Ven, J. T. Vogelstein, F. Wang, R. Weiss, A. Yanguas-Gil, X. Zou, H. Siegelmann, Biological underpinnings for lifelong learning machines. *Nat. Mach. Intell.* **4**, 196–210 (2022).
40. M. Gelbrich, W. Römisch, Numerical solution of stochastic differential equations (Peter E. Kloeden and Eckhard Platen). *SIAM Rev.* **37**, 272–275 (1995).
41. H. Khalil, *Nonlinear Systems* (Prentice Hall, 2002).
42. D. Horn, J. Weyers, Hypercubic structures in orthogonal Hopfield models. *Phys. Rev. A* **36**, 4968–4974 (1987).
43. R. A. Horn, C. R. Johnson, *Matrix Analysis* (Cambridge Univ. Press, 2012).
44. P. H. Baxendale, S. V. Lototsky, *Stochastic Differential Equations: Theory and Applications: A Volume in Honor of Professor Boris L. Rozovskii* (World Scientific, 2007).
45. G. A. Pavliotis, *Stochastic Processes and Applications: Diffusion Processes, the Fokker-Planck and Langevin Equations* (Springer, 2014).
46. R. Jordan, D. Kinderlehrer, F. Otto, Free energy and the Fokker-Planck equation. *Phys. D: Nonlinear Phenom.* **107**, 265–271 (1997).
47. R. Jordan, D. Kinderlehrer, F. Otto, The variational formulation of the Fokker-Planck equation. *SIAM J. Math. Anal.* **29**, doi.org/10.1137/S0036141096303359 (1998).
48. F. Blanchini, S. Miani, *Set-Theoretic Methods in Control* (Springer International Publishing, 2015).

Acknowledgments

Funding: This work was, in part, supported by AFOSR project FA9550-21-1-0203 (F.B.) and by the grant Next Generation EU C96E22000350007 (S.B.). **Author contributions:** Writing—original draft: S.B., G.B., F.B., and S.Z. Writing—review and editing: S.B., G.B., F.B., and S.Z. Conceptualization: S.B., G.B., F.B., and S.Z. Investigation: S.B., G.B., and S.Z. Methodology: S.B., G.B., F.B., and S.Z. Formal analysis: S.B., G.B., F.B., and S.Z. Data curation: S.B. Validation: S.B. Software: S.B. Visualization: S.B., G.B., and F.B. Supervision: G.B., F.B., and S.Z. Funding acquisition: F.B. and S.Z. Project administration: F.B. Resources: S.Z. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** The code used in this manuscript is publicly available at <https://doi.org/10.5281/zenodo.14873595>. All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials.

Submitted 18 November 2024

Accepted 24 March 2025

Published 23 April 2025

10.1126/sciadv.adu6991

Input-driven dynamics for robust memory retrieval in Hopfield networks

Simone Betteti, Giacomo Baggio, Francesco Bullo, and Sandro Zampieri

Sci. Adv. **11** (17), eadu6991. DOI: 10.1126/sciadv.adu6991

View the article online

<https://www.science.org/doi/10.1126/sciadv.adu6991>

Permissions

<https://www.science.org/help/reprints-and-permissions>

Use of this article is subject to the [Terms of service](#)

Science Advances (ISSN 2375-2548) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science Advances* is a registered trademark of AAAS.

Copyright © 2025 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC).